

Landslide Debris-Flow Prediction using Ensemble and Non-Ensemble Machine-Learning Methods

Praveen Kumar¹, Priyanka Sihag¹, Ankush Pathania¹, Shubham Agarwal¹,
Naresh Mali², Pratik Chaturvedi³, Ravinder Singh⁴,
K.V. Uday², and Varun Dutt¹

¹Applied Cognitive Science Lab, Indian Institute of Technology Mandi, Himachal Pradesh, India-175005

²Geohazard Studies Laboratory, Indian Institute of Technology Mandi, Himachal Pradesh, India-175005

³Defence Terrain Research Laboratory, Defence Research and Development Organization (DRDO), New Delhi, India-110054

⁴National Disaster Management Authority (NDMA), New Delhi, India-110029
bluecodeindia@gmail.com, priyankasihag8993@gmail.com,
ankushpathania.ap@gmail.com, shubhamagarwal223@gmail.com,
malinareshmudhiraj@gmail.com, prateek@dttrl.drdo.in,
ravinder.ndma@gmail.com, uday@iitmandi.ac.in,
varun@iitmandi.ac.in

Abstract. Landslides and associated soil movements (debris-flow) are the common natural calamities in the hilly regions. In particular, Tangni in Uttarakhand state between Pipalkoti and Joshimath has experienced a number of landslides in the recent past. Prior research has used certain machine-learning (ML) algorithms to predict landslides. However, a comparison of ensemble and non-ensemble ML algorithms for debris-flow predictions has not been undertaken. In this paper, we use ensemble and non-ensemble machine-learning (ML) algorithms to predict debris-flow at the Tangni landslide. Non-ensemble algorithms (Sequential Minimal Optimization (SMO), and Autoregression) and ensemble algorithms (Random Forest, Bagging, Stacking, and Voting) involving the non-ensemble algorithms were used to predict weekly debris-flow at Tangni between 2013 and 2014. Result revealed that the ensemble algorithms (Bagging, Stacking, and Random Forest) performed better compared to non-ensemble algorithms. We highlight the implications of predicting debris-flow ahead of time in landslide-prone areas in the world.

Keywords: Landslides, Sequential Minimal Optimization (SMO), Autoregression, Random Forest, Bagging, Stacking, Voting.

1 Introduction

Landslides and soil-movements (debris-flow) are caused due to the rain, mostly in the monsoon season in hilly areas [1]. These landslides and debris-flow are

natural hazards that often happen without warning and cause massive loss of property and life across the world [2]. Landslides are a major problem in India with a staggering 11,000 deaths over the past 12 years in the country [3]. Machine-learning (ML) algorithms, which allow prediction of future outcomes based upon historical data, hold the key for timely alerting people about debris-flow and impending landslides [4]. As mentioned above, the landslide debris-flow causes damages to lives and property [5], and ML algorithms could be utilized in predicting debris-flow in landslide-prone areas [6]. Here, ML algorithms could learn patterns in data collected by sensor systems that are installed in different landslide-prone locations [7].

Although there is a large class of ML algorithms that have been proposed for making predictions about debris-flow [15-23], a comparison of these algorithms with ensemble versions of existing algorithms have been less investigated. In this paper, we use several ensemble and non-ensemble ML algorithms for predicting debris-flow ahead of time. For example, different non-ensemble algorithms (SMO [8], Autoregression (AR) [9]), and ensemble algorithms (Random Forest (RF) [10], Bagging [11], Stacking [12], and Voting [13]) have been used in the past for time series forecasting. SMO is a variant of the support-vector machine (SVM) algorithm that optimizes the training of SVM for regression problems. Autoregression is mostly used for predicting the values of variables based upon the prior values of the same variables. Random Forest, an ensemble algorithm, produces its predictions using a collection of decision trees with different feature sets. Bagging, bootstrapping and aggregation, uses subsamples from a dataset with replacement and creates predictive models from training on different subsamples. Stacking is combining the predictions of multiple models into a single model that makes the final prediction. Voting algorithm is used for combining the prediction of different ML algorithms by averaging their predictions to generate the time-series forecasts.

The primary goal of this paper is to evaluate non-ensemble ML algorithms (SMO and Autoregression) with ensemble versions of these algorithms (Random Forest, Bagging, Stacking, and Voting) to predict debris-flow over time at a real-world landslide location. Specifically, we use weekly debris-flow sensor data collected at the Tangni landslide between 2013 and 2014 and use it to predict debris-flow one-week ahead of time.

In what follows, we first detail the background literature concerning the use of ML algorithms for soil-movement predictions. Next, we detail the study area and data used for our time-series forecasting. Then, we detail different ensemble and non-ensemble methods that we developed on debris-flow data. Finally, we detail our results and discuss the main implications of our results for debris-flow time-series predictions in the real world.

2 Background

Prior research has explored landslide susceptibility mapping and debris-flow prediction through ML algorithms [14-22]. Some researcher has used support-vector-

based models for debris-flow predictions [14–18]. For example, reference [14] predicted the slope displacement in the Three Gorges Reservoir, China, using a Particle Swarm Optimization and Support Vector Machine (PSO–SVM) coupling model. Results revealed that the proposed PSO–SVM model could represent the relationship between the causal factors and the cyclic slope displacements. Similarly, reference [15] have found that the Gaussian process performed better than the simple artificial neural network (ANN) models, relevance vector machine, and support vector machine.

Furthermore, reference [16] used a case-study of landslides in the Ecuadorian Andes and compared the prediction power of support vector machines, logistic regression, and bootstrap-aggregated classification trees (Bagging and Double-Bagging). Results revealed that logistic regression with stepwise backward variable selection produced the lowest error rates and provided the best generalization capabilities. Next, reference [17] compared a Least Square Support Vector Machine (LSSVM) model boosted with Genetic Algorithm, namely GA-LSSVM, with a Double Exponential Smoothing (DES) and LSSVM model to empirically forecast landslide displacements. Some researchers have also found that a Support Vector Machine (SVM) regression predicted debris-flow in Baishuihe landslide in Three Gorges Reservoir Area, China, with a small error [18].

Reference [19] have used an autoregressive (AR) model and a detrended fluctuation analysis (DFA) method to model debris-flow. The coefficient and variance of AR and the scaling exponent of DFA were estimated using a slide window. Similarly, reference [20] used autoregressive moving average time-series models to analyze the autocorrelation of landslide triggering factors.

Some researchers have also used tree-based models for predicting debris-flow [21]. For example, reference [21] showed that the Random Forest model was capable of predicting the evolution of daily slope displacements over a 30-day period.

Reference [22] used a hybrid bagging-based method for the prediction of landslides at the district of Mu Cang Chai, Vietnam. In this study, 248 past landslides and fifteen geo-environmental factors were considered for the model construction. Based on the AUC values, bagging outperformed the SVM and NBT models. Thus, ensemble methods like bagging may provide promising methods for landslide debris-flow predictions.

3 Study Area

The study was performed in on the Tangni landslide in Chamoli district of Uttarakhand, India. The study area covers an area of 0.72 km². It is located on the northern Himalayan region at latitudes 30° 27' 54.3" N and longitudes 79° 27' 26.3" E, at an altitude of 1450 meter (Fig. 1A and 1B). As seen in Fig. 1B, the landslide is located on National Highway 58, which connects Ghaziabad in Uttar Pradesh near New Delhi with Badrinath and Mana Pass in Uttarakhand. The geology of this area consists of slate and dolomite rocks [21]. The landslide slope is 30° above the road level and 42° below the road level. There have been

several prior landslides in this area causing roadblocks and economic losses to tourism [23].



Fig. 1. (A) Location of the study area. (B) The Tangni landslide on Google Maps.

Data was collected from the Tangni landslide at a daily scale between 1st July 2012 and 1st July 2014 across five different boreholes (BH). These five boreholes are represented by different colors in Fig. 1B (red – borehole 1, green – borehole 2, yellow – borehole 3, blue – borehole 4, and pink – borehole 5). Each borehole contained five sensors at different depths (3m, 6m, 9m, 12m, and 15m). Data from some of these five boreholes was used for evaluating different ensemble and non-ensemble machine-learning methods.

4 Methodology

4.1 Data Pre-Processing

Data from Tangni landslide in Chamoli, India was obtained from the Defence Terrain Research Laboratory, Defence Research and Development Organization. The monitoring system in each of the five boreholes at the Tangni landslide contained inclinometer sensors at different depths (3m, 6m, 9m, 12m, and 15m). These sensors measured tilt in mm per m units (essentially the angle the inclinometer tilts). Each inclinometer sensor was a 0.5-meter long sensor that was installed vertically at different depths in a borehole. The monitoring system at Tangni landslide had five sensors per borehole across five boreholes. Thus, in total there are 25 sensors across 5 boreholes. Fig. 2 shows the inclinometer sensor installed in its casing at a certain depth. As shown in Fig. 2, if there was a tilting movement (θ) of the inclinometer of length L , then the horizontal displacement in the tilting direction was $L \sin \theta$. For better understanding, we converted the displacement in mm per m units into a θ angle (degrees), where 1mm/m displacement equalled 0.05729° . First, we calculated the relative tilt angle of each sensor from its initial reading at the time of installation. Second, we chose only those sensors from each borehole that gave the maximum average tilt angle over a two-year period. Thus, the data was reduced to five time-series, where each time-series represented the relative tilt per borehole from the sensor that moved the most in the borehole across the two-year period. As the daily data was sparse, we averaged the tilt over weeks to yield 78 weeks of average

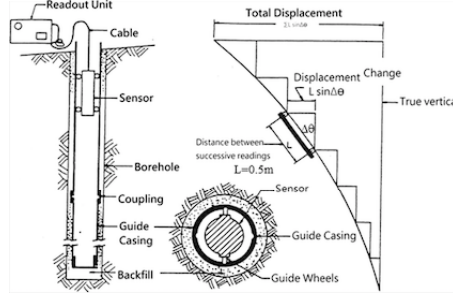


Fig. 2. Inclinometer sensor installed in its casing at a certain depth.

tilt data per time-series. Fig. 3A-3E represent the average relative tilt per week from five sensors across five boreholes (one sensor per borehole) that caused the maximum average tilt across 78 weeks. These five time-series were used to compare the ensemble and non-ensemble methods.

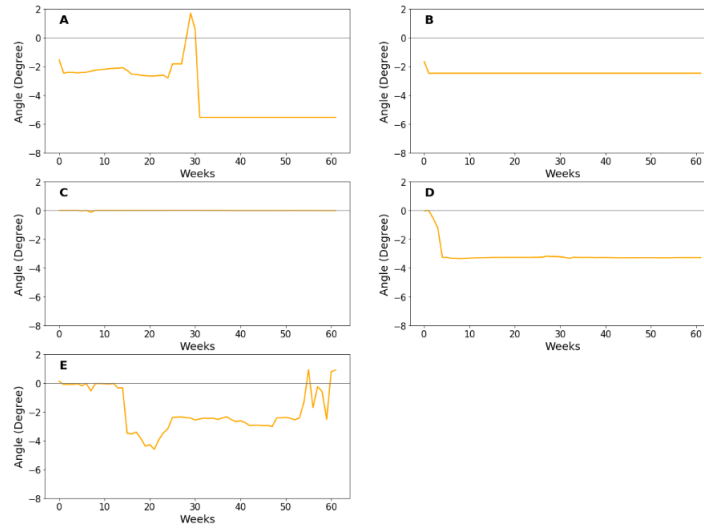


Fig. 3. Average tilt angle in degrees across five sensors (one per borehole). (A) Borehole 1 and 3m sensor. (B) Borehole 2 and 12m sensor. (C) Borehole 3 and 6m sensor. (D) Borehole 4 and 15m sensor. (E) Borehole 5 and 15m sensor.

By convention, a negative tilt angle was a downhill motion and a positive tilt angle was an uphill motion. As seen in Fig. 3C, a downhill motion starts from -0.11° in the 73^{rd} week and suddenly becomes larger (-4.4°) in the last four weeks. The data was split in an 80:20 ratio (sixty-two weeks for training and the last sixteen weeks for testing) across different machine learning algorithms.

4.2 Sequential Minimal Optimization

John Platt proposed the sequential minimal optimization (SMO) in 1998 [8]. It is a widely-used algorithm for solving a quadratic programming (QP) problem that arises during the training of support vector machines. The goal of the SMO algorithm is to return alpha parameters (Lagrange multipliers) that satisfy the following constraint optimization problem:

$$\min_{\alpha} \sum_i \sum_j \alpha_i \alpha_j y_{ij} X_i \cdot K(X_i, X_j) - \sum \alpha_i \quad (1)$$

For a Kernel function:

$$K(X_i, X_j) = \phi(X_i) \cdot \phi(X_j) \quad (2)$$

and

$$s.t. \sum_i \alpha_i y_i = 0, \alpha_i \in [0, C] \quad (3)$$

4.3 Autoregression

Autoregression (AR) is a time series model that uses observations from previous time steps as input to a regression equation to predict the value at the next time step. This technique can be used on a time-series where input variables are taken as observations at previous time steps, called lag variables. For example, we can predict the value for the next time step (t+1) given the observations at the last two time-steps (t-1 and t-2). As a regression model, this would look as follows (X_{t+1} is the movement to be predicted using the movements (X_{t-1} and X_{t-2})) :

$$X_{t+1} = \beta_0 + \beta_1 \cdot X_{t-1} + \beta_2 \cdot X_{t-2} \quad (4)$$

4.4 Random Forest

Random Forest algorithm was developed by Leo Breiman and Adele Cutler [9]. Random Forest or random decision forests are an ensemble learning method for classification and regression. At the time of training, the Random Forest algorithm output is the majority class (classification) or the value that is the mean of the prediction of individual trees (regression). By aggregation, the Random Forest algorithm corrects the problem of over fitting in decision trees [10].

4.5 Voting

Voting is perhaps the simplest ensemble algorithm and it is often very effective [13]. Voting can be used for classification or regression problems. Voting is one of the ensemble methods that allows to improve accuracy by means of combining several base models. There are several possible ways to organize voting procedure: majority voting, average of probabilities, median of probabilities, etc. In this paper, we consider the average of probabilities model as it is the default assumption in several machine-learning tools.

4.6 Bagging

Bagging is a machine learning ensemble Meta algorithm designed to improve the stability and accuracy of machine learning algorithms used in statistical classification and regression [11]. Bagging uses subsamples from the dataset with replacement and trains the predictive model on these subsamples. The final output model is averaged across all models for the better result.

4.7 Stacking

Stacking is an ensemble learning technique that combines multiple classification or regression models via a meta-classifier or a meta-regressor [12]. In Stacking, the algorithm takes the outputs of sub-modes as inputs and attempts to learn how to best combine the input predictions to make a better output prediction.

4.8 Optimization of Model Parameters

Sequential Minimal Optimization (SMO). SMO algorithm has two parameters. The first parameter is the complexity parameter (C) that is used to build a “hyperplane” between two classes, which are used for classification, regression, or other tasks. The C parameter controls how many instances are used as “support vectors” to draw a linear separation boundary in the transformed Euclidean feature space. The second parameter of the SMO algorithm is an exponent (E) of the kernel function. We varied the C and E parameters in SMO as per the following: C=0, 1 and E=1, 2, 3, 4 for polynomial kernel; C=0, 1 and E=1, 2 for normalized polynomial kernel; and, C=0 and E = 1 for RBF kernel.

Random Forest. In Random Forest, a key parameter is the number of features (nF) to consider in each split point. In this paper, we varied the nF parameter between 0 and 10 to calibrate the Random Forest model.

Autoregression. This algorithm has parameters corresponding to the beta coefficients ($\beta_0, \beta_1, \beta_2, \dots, \beta_n$) and the last n lag terms (t-1, ..., t-n). We varied ridge parameter value from 1 to 50 with step size=10 [24]. We tried the two options for attribute selection, M5 and greedy, that is known as the Akaike Information Criterion (AIC) to select features for the linear regression.

Voting. In voting, we used SMO, Autoregression, and Random Forest as the sub models. A key parameter in voting is how the predictions of the sub models are combined. In this paper, we have assumed the combination rule to be average of the probabilities, minimum probability, maximum probability, and median.

Stacking. In Stacking, we tried SMO, Autoregression, and Random Forest as the sub models. Furthermore, IBk, K Star, LWL were used as the meta-learner. The number of folds used for cross-validation (numFold) were varied between 10 to 100 with step size 10 to find best value of parameters.

Bagging. In Bagging, we used SMO, Autoregression, and Random Forest one at a time as a sub model. Furthermore, the number of bags (I) were varied between 10 and 150 with step size of 5 to find this parameter's best value.

5 Results

Each algorithm was calibrated to each of the five time-series independently. Table 1 shows the root-mean squared error (RMSE) results of applying ensemble and non-ensemble algorithms, AR, SMO, Random Forest, Bagging Voting and Stacking, on the training data across the five boreholes. As can be seen in Table 1, the Random Forest and Voting (ensemble) algorithms performed the best and second best and better compared to the other algorithms.

Table 1. The RMSE of different algorithms in the training dataset.

Root-Mean Squared Error (RMSE) in angle °							
Rank	Algorithm	Borehole 1 3m	Borehole 2 12m	Borehole 3 6m	Borehole 4 15m	Borehole 5 15m	Average
5	SMO	0.95	0.00	0.00	0.07	0.83	0.37
6	Autoregression	1.02	0.01	0.00	0.12	0.85	0.40
1	Random Forest	0.39	0.00	0.00	0.01	0.31	0.14
3	Voting	0.96	0.01	0.00	0.11	0.52	0.32
4	Bagging	0.94	0.00	0.00	0.01	0.70	0.33
2	Stacking	0.60	0.00	0.00	0.02	0.70	0.26

Table 2 shows the optimized values of different parameters of ensemble and non-ensemble algorithms. For example, in Random Forest algorithm, the Bags (I) = 50 and Number of Features (nF) = 5. Similarly, Bagging used I = 5.

Table 2. Best set of parameters.

Algorithm	Parameters
SMO	C=5, E=1, Polynomial Kernel
Autoregressive	Ridge=30, Attribute Selection Method=M5
Random Forest	Bags (I)=50, Number of Features (nF)=5
Voting	Combination Rule=[Minimum of Probabilities]
Bagging	Bags (I)=25, Classifiers=[SMO]
Stacking	Meta Classifier=[SMO], numFold=50

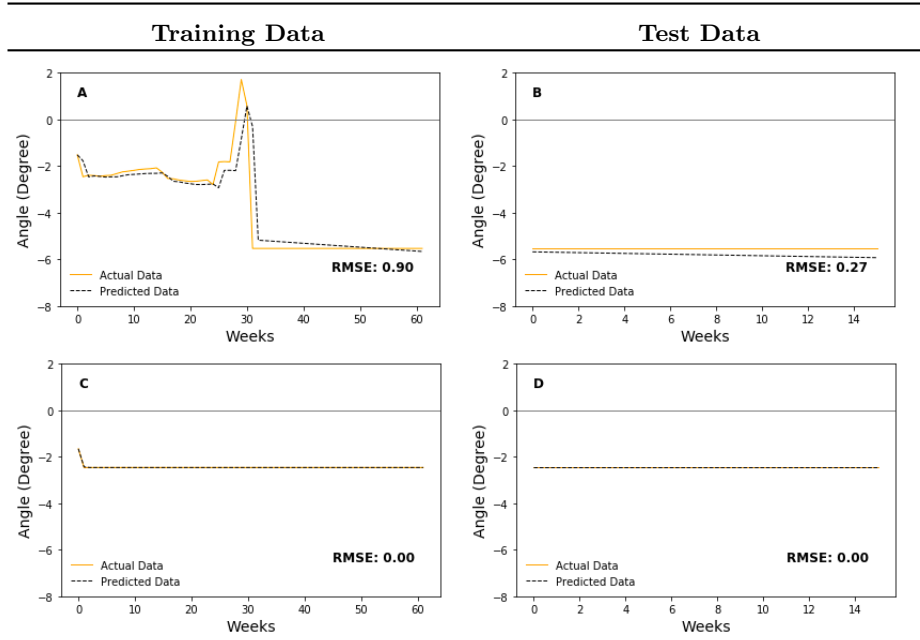
Table 3 shows the RMSEs from different models across different boreholes in the last 16-weeks of test data. As can be seen in the table, Bagging, Random

Table 3. The RMSE of different algorithms in the testing dataset.

Root-Mean Squared Error (RMSE) in angle °							
Rank	Algorithm	Borehole 1 3m	Borehole 2 12m	Borehole 3 6m	Borehole 4 15m	Borehole 5 15m	Average
4	SMO	0.03	0.00	1.36	0.06	1.26	0.54
5	Autoregression	0.25	0.01	1.45	0.16	1.12	0.59
3	Random Forest	0.00	0.00	1.54	0.09	1.00	0.52
6	Voting	0.03	0.18	0.58	1.23	1.29	0.66
1	Bagging	0.06	0.00	0.73	0.08	1.30	0.43
2	Stacking	0.00	0.00	1.55	0.09	0.97	0.52

Forest, and Voting performed the best, second best, and third best among all algorithms.

Fig. 4. shows the fits of the Bagging algorithm to the time-series data across the five boreholes in the training and test datasets. Overall, these results are reasonably good with very small RMSE values.



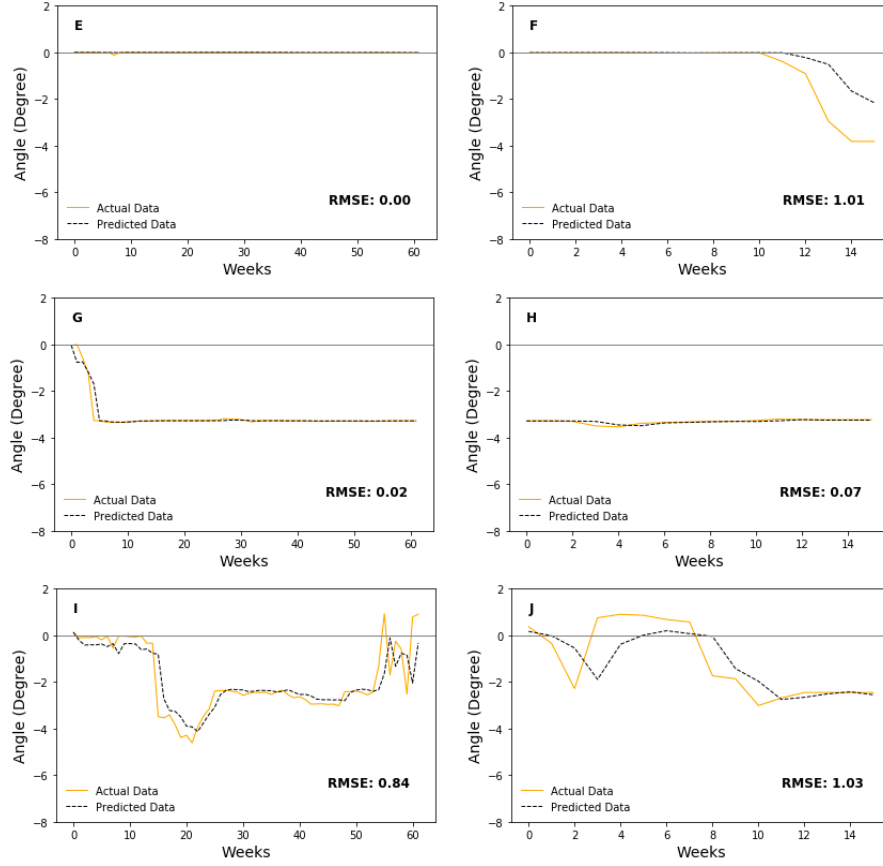


Fig. 4. Relative angle (in degree) over training data (62 weeks) and test data (16 weeks) from the best performing Bagging algorithm. (A) and (B): borehole 1, 3m depth. (C) and (D): borehole 2, 12m depth. (E) and (F): borehole 3, 6m depth. (G) and (H): borehole 4, 15m depth. (I) and (J): borehole 5, 15m depth.

6 Discussion and Conclusions

A focus of machine-learning (ML) algorithms could be the prediction of debris-flow in advance to timely warn people about impending landslides. In this work, we applied both ensemble and non-ensemble ML algorithms on weekly debris-flow data from the Tangni landslide in Chamoli, India. All models were calibrated on the first 80% of data and tested on the last 20% of data. All models could generate the debris-flow predictions in the following week given the history of movements in prior weeks. Our results revealed that the ensemble algorithms performed better compared to non-ensemble algorithms during both model training and test.

First, we found that the Bagging algorithm performed the best among all algorithms. A likely reason for this result could be that the Bagging algorithm includes two operations as part of its functioning: Bootstrap and Aggregation. Bootstrap involves estimating means from multiple random samples of data with replacement. This mean estimation may be useful especially in cases where we have limited amount of data. In addition, the aggregation involved many bags ($= 25$), which likely yielded superior performance.

Second, we found that the Random Forest algorithm performed the second best among all algorithms. A likely reason for this result could be that the Random Forest algorithm is an ensemble of several decision trees with varying number of features. Also, this algorithm builds upon the Bagging algorithm and includes features like different bag sizes.

In this paper, we were able to show that both ensemble and non-ensemble algorithms may be used for debris-flow predictions. However, as part of our future research, we plan to extend these analyses to other neural-network-based methods including the use of both artificial neural networks as well as recurrent neural networks (e.g., long short-term memory models). Also, we plan to combine these methods using different ensembling techniques like bagging. Some of these ideas form the immediate next steps in our program on debris-flow predictions using ML techniques.

Acknowledgments The project was supported from grants (awards: IITM/NDMA/VD/184, IITM/DRDO-DTRL/VD/179, and IITM/DCoN/VD/204) to Varun Dutt. We are also grateful to Indian Institute of Technology Mandi for providing computational resources for this project.

References

1. Mathew, D. Babu, S. Kundu, K. Kumar and C. Pant.: Integrating intensity–duration-based rainfall threshold and antecedent rainfall-based probability estimate towards generating early warning for rainfall-induced landslides in parts of the Garhwal Himalaya, India”, *Landslides*, vol. 11(4), pp. 575–588 (2013).
2. Melanie J. Froude and David N. Petley, ”Global fatal landslide occurrence from 2004 to 2016”, *Nat. Hazards Earth Syst. Sci.*, 18, 2161–2181, 2018.
3. The Weather Channel (October 2018). Why India Needs to be Concerned About Landslides. <https://weather.com/en-IN/india/science/news/2018-10-18-why-india-needs-to-be-concerned-about-landslides>, last accessed April 7, 2019.
4. Chaturvedi, P., Srivastava, S., Kaur, P. B.: Landslide EarlyWarning System Development Using Statistical Analysis of Sensors’ Data at Tangni Landslide, Uttarakhand, India. In: *Sixth International Conference on Soft Computing for Problem Solving* (2017), AISC, volume 547, pp. 259–270. Springer, Singapore (2017).
5. Parkash, S.: Historical Records of Socio-economically Significant Landslides in India. *Journal of South Asia Disaster Studies* 4(2), 177–204 (2011).
6. Du, J., Yin, K., Lacasse, S.: Displacement prediction in colluvial landslides, three Gorges reservoir, China. *Landslides*, vol. 10(2), pp. 203–218 (2013).

7. Mali, N., Chaturvedi, P., Dutt, V., Kala, V. U.: Training of Sensors for Early Warning System of Rainfall-Induced Landslides. In *Recent Advances in Geo-Environmental Engineering, Geomechanics and Geotechnics, and Geohazards*, pp. 449-452. Springer, Cham (2019).
8. Platt, J. (April 1998). Sequential Minimal Optimization: A Fast Algorithm for Training Support Vector Machines. <https://www.microsoft.com/en-us/research/publication/sequential-minimal-optimization-a-fast-algorithm-for-training-support-vector-machines/>, last accessed April 7, 2019.
9. Ehrmann, M., Ellison, M., Valla, N.: Regime-dependent impulse response functions in a Markov-switching vector autoregression model. *Economics Letters*, vol. 78(3), pp. 295-299 (2003).
10. Breiman, L.: Random forest, *Machine Learning*, vol. 45(1), pp. 5-32 (2001).
11. Breiman, L.: Bagging predictors. *Machine Learning*, vol. 24(2), pp. 123-140 (1996).
12. Ting, K.M., Witten, I.H.: Stacked generalization: when does it work? Hamilton, New Zealand: University of Waikato, Working paper 97/03, (1997).
13. Thornton, C., Hutter, F., Hoos, H. H., Leyton-Brown, K. . Auto-WEKA: Combined selection and hyperparameter optimization of classification algorithms. In *Proceedings of the 19th ACM SIGKDD international conference on Knowledge discovery and data mining*. ACM. pp. 847-855, (2013).
14. Zhou, C., Yin, K., Cao, Y., Ahmed, B.: Application of time series analysis and PSO-SVM model in predicting the Bazimen landslide in the Three Gorges Reservoir, China. *Engineering geology*, vol. 204, pp. 108-120, (2016).
15. Liu, Z., Shao, J., Xu, W., Chen, H., Shi, C.: Comparison on landslide non-linear displacement analysis and prediction with computational intelligence approaches. *Landslides*, vol. 11(5), pp. 889-896 (2014).
16. Brenning, A.: Spatial prediction models for landslide hazards: review, comparison and evaluation. *Natural Hazards and Earth System Science*, v.5(6), p853-862, 2005.
17. Zhu, X., Xu, Q., Tang, M., Nie, W., Ma, S.: Comparison of two optimized machine learning models for predicting displacement of rainfall-induced landslide: A case study in Sichuan Province, China. *Engineering geology*, vol. 218, pp. 213-222, 2017.
18. Zhu, C. H., Hu, G. D.: Time series prediction of landslide displacement using SVM model: application to Baishuihe landslide in Three Gorges reservoir area, China. In *Applied Mechanics and Materials*, vol. 239, pp. 1413-1420, (2013).
19. Pu, F., Ma, J., Zeng, D., Xu, X., Chen, N.: Early warning of abrupt displacement change at the Yemaomian landslide of the Three Gorge Region, China. *Natural Hazards Review*, vol. 16(4), pp. 04015004 (2015).
20. Li, H., Xu, Q., He, Y., Deng, J.: Prediction of landslide displacement with an ensemble-based extreme learning machine and copula models. *Landslides*, vol. 15(10), pp. 2047-2059 (2018).
21. Krkač, M., Špoljarić, D., Bernat, S., Arbanas, S. M.: Method for prediction of landslide movements based on random forests. *Landslides*, vol. 14(3), pp. 947-960, (2017).
22. Pham, B. T., Bui, D. T., Prakash, I.: Bagging based Support Vector Machines for spatial prediction of landslides. *Environmental earth sciences*, vol. 77(4), pp. 146, (2018).
23. India News (2013, August 13). Landslides near Badrinath in Uttarakhand. <https://www.indiatvnews.com/news/india/landslides-near-badrinath-in-uttarakhand-26296.html>, last accessed April 7, (2019).
24. Dorugade, A. V.: New ridge parameters for ridge regression. *Journal of the Association of Arab Universities for Basic and Applied Sciences*, vol. 15(1), pp. 94-99, (2014).